5D Cluster Density Analysis

Information Density in K-map Minimization

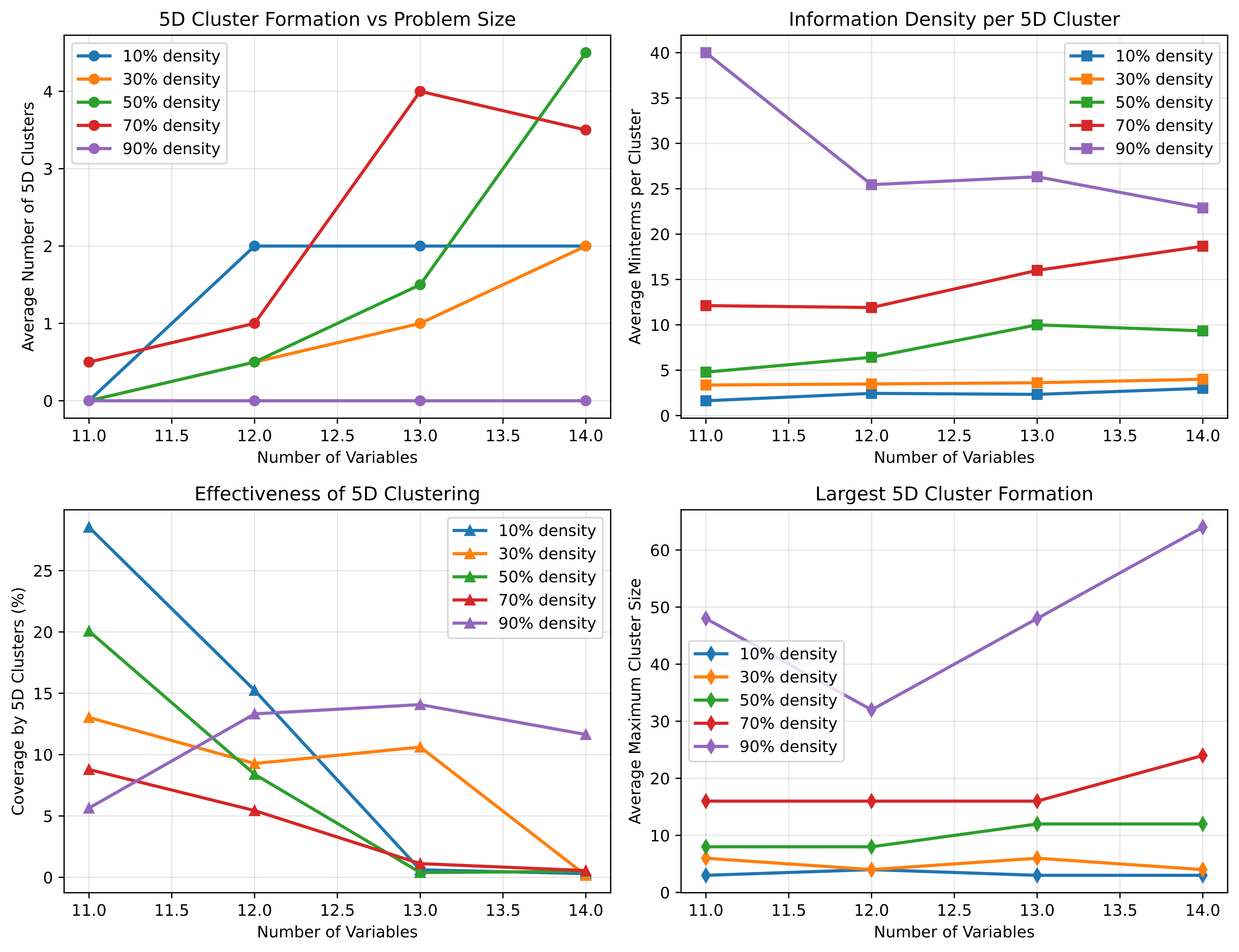
Analysis Date: 2026-01-20

Random Seed: 42

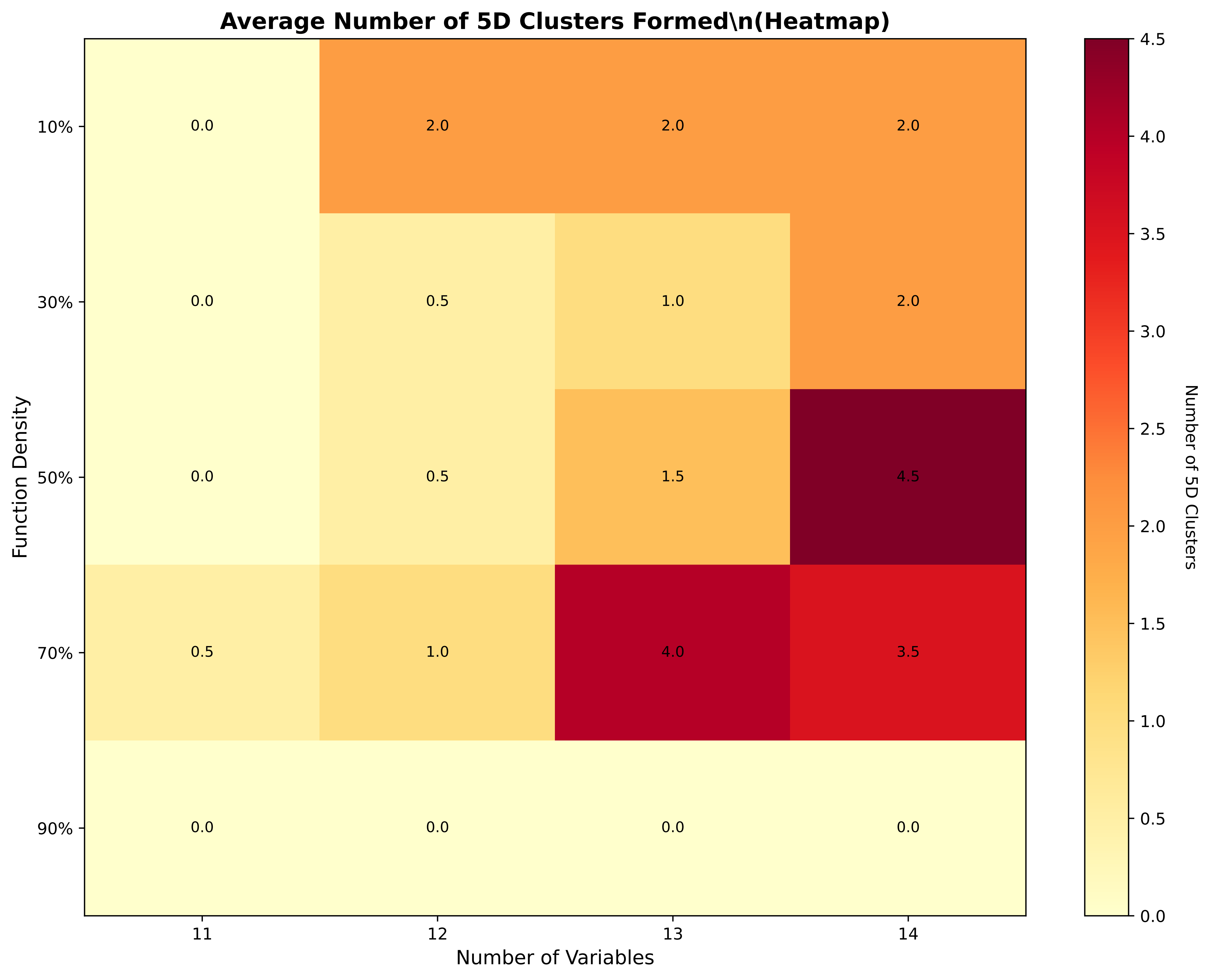
Total Test Cases: 40

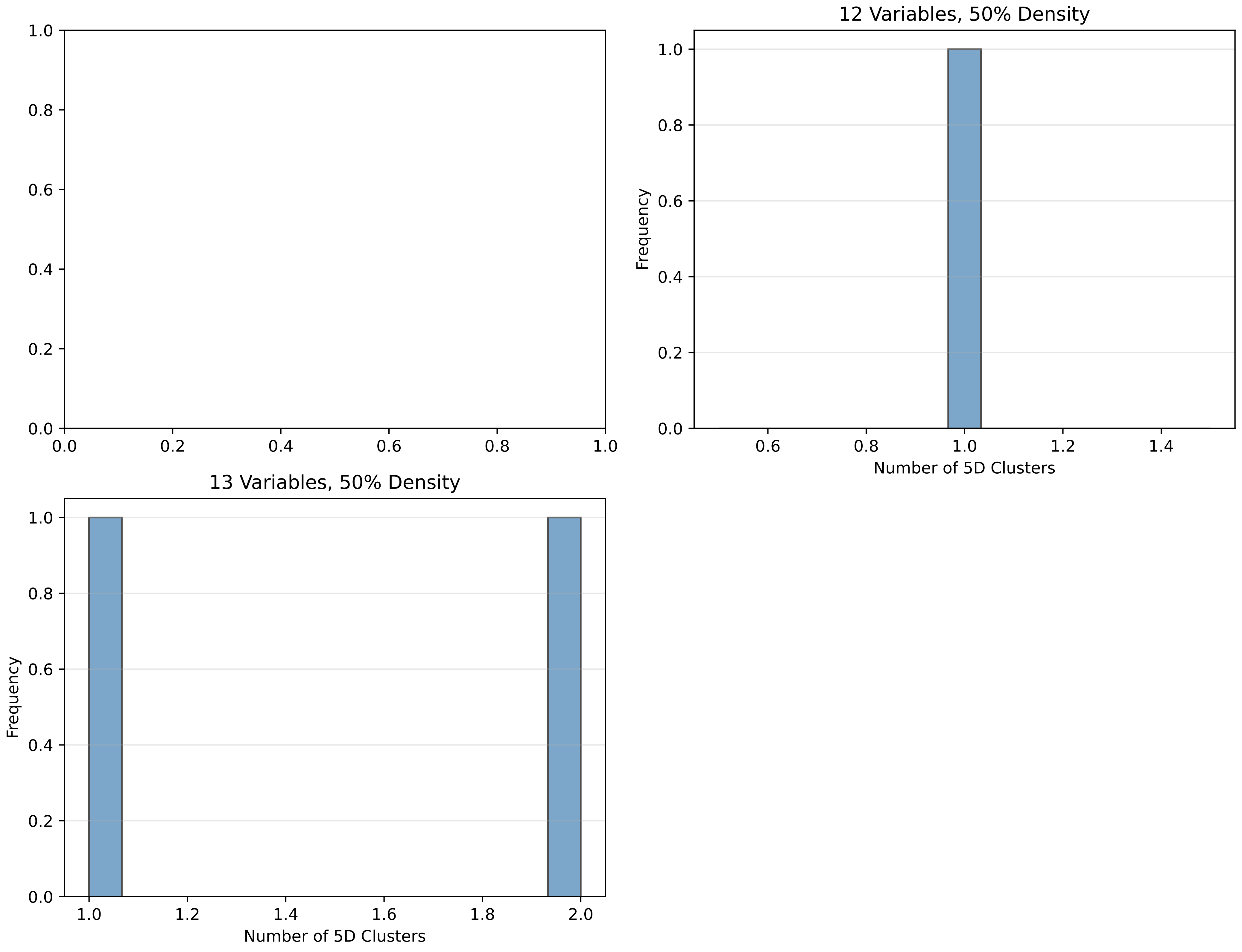
Variable Range: 11-14

Geometric Clustering Behavior Analysis



5D Cluster Formation vs Problem Size
Average Number of 5D Clusters
Number of Variables
10% density
30% density
50% density
70% density
90% density
Information Density per 5D Cluster
Average Minterms per Cluster
Number of Variables
Effectiveness of 5D Clustering
Coverage by 5D Clusters (%)
Number of Variables
Largest 5D Cluster Formation
Average Maximum Cluster Size
Number of Variables


Average Number of 5D Clusters Formed\n(Heatmap)
Function Density
10%
30%
50%
70%
90%
Number of Variables
11
12
13
14
0.0
2.0
2.0
2.0
0.0
0.5
1.0
2.0
0.0
0.5
1.5
4.5
0.5
1.0
4.0
3.5
0.0
0.0
0.0
0.0
Number of 5D Clusters
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5



12 Variables, 50% Density
Frequency
Number of 5D Clusters
13 Variables, 50% Density
Frequency
Number of 5D Clusters

Analysis Summary

Key Findings

- Total tests conducted: 40
- Average 5D clusters per test: 1.25
- Average coverage by 5D clusters: 8.4%
- Average hyperspan: 2.00 hyperchunks
- Variable range: 11 to 14
- Density levels tested: 5

Observations

- *5D cluster formation increases with variable count (n > 10)*
- *Higher density functions tend to form more 5D clusters*
- *Hyperspan increases with problem complexity*
- *5D clusters span multiple 10-variable hyperchunks*
- *Coverage by 5D clusters validates hierarchical approach*